

Target Region: CENTRAL INDIA	Daily Volume: 250 GB/day	BI Users: 50 Users	Workload Mix: BI_ONLY
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Executive Architecture Recommendation

Enterprise Platform Decision & Sizing Saturated Matrix

Data Confidence:  75%

RECOMMENDED PLATFORM

Microsoft Fabric

96% MATCH

Based on your input of 250 GB/day processed, 50 peak BI users, and sql powerbi team skillset, Microsoft Fabric is the recommended architecture platform for your enterprise workload.

\$ COST / MO

₹4,88,750 – ₹9,69,000

SKU SIZING

F32

CONFIDENCE

75%

MIGRATION

Low

DURATION

3–6 Weeks

POTENTIAL SAVINGS

18% - 35%

SCANNABLE DECISION BADGES

✓ Native Direct Lake

✓ Zero Cluster Management

✓ Unified M365 SaaS

✓ Single Capacity Billing

⚠ Fixed Memory Limits

⚠ Less Mature MLOps

⚠ Potential Concurrency Queuing

✓ WHY MICROSOFT FABRIC WAS SELECTED:

- Your BI-only workload and SQL/Power BI skillset map directly to Fabric's unified SaaS experience and Direct Lake mode.
- Fabric eliminates the need for cluster management, workspace setup, and complex ETL pipelines.
- Estimated monthly cost range (₹4,88,750 – ₹9,69,000) delivers lower TCO without wasted spend on unused ML infrastructure.

 WHY NON-SELECTED PLATFORM WAS NOT CHOSEN:

- Databricks introduces unnecessary cluster management overhead and VM infrastructure complexity for pure BI reporting.
- Databricks lacks native zero-copy Direct Lake mode, requiring Power BI queries to keep SQL Warehouses running.
- Higher learning curve for SQL & Power BI developers compared to Fabric's native SaaS environment.

 Migration Roadmap & Timeline Estimate

Enterprise Transition Assessment

MIGRATION DIFFICULTY

Low Complexity



ESTIMATED DURATION

3–6 Weeks

~320 Total Engineering Hours

RISK LEVEL

 **Low Risk**

Standard automated ingestion & schema migration



Microsoft Fabric

Fit Score: 96/100

Unified SaaS Analytics Platform

Data Confidence: 75%

₹4,88,750 – ₹9,69,000

Estimated Monthly Cost Range

₹58,65,000 – ₹1,16,28,000 / yr

\$ Cost Itemization Breakdown

Monthly

Capacity Compute (CU)	₹3,91,000
OneLake Storage	₹14,663
SQL Data Warehouse	₹85,000
Power BI Integration	Included in F-SKU

 Suggested Sizing: F32

PRIMARY SIZING DRIVER

Peak concurrent BI users (50 users → ~28 CUs floor)

Key Sizing Assumptions:

- Fabric Capacity pay-as-you-go rate in Central India (Pune) (~₹4,75,988/mo base for F32).
- 730 hours/month capacity availability with dynamic smoothing and bursting enabled.
- OneLake storage estimated at 7500 GB (₹14,663/mo).



Azure Databricks

Fit Score: 87/100

Managed Spark & Lakehouse Engine

Data Confidence: 75%

₹55,250 – ₹80,750

Estimated Monthly Cost Range

₹6,63,000 – ₹9,69,000 / yr

\$ Cost Itemization Breakdown

Monthly

Databricks Compute (DBUs)	₹34,000
ADLS Gen2 Storage	₹12,750
Azure VM Infrastructure	₹17,000
Unity Catalog Governance	Unity Catalog Included

 Suggested Sizing: 4-Node SQL Warehouse + 2-Node Jobs Cluster

PRIMARY SIZING DRIVER

SQL Warehouse serverless compute (4 nodes for 50 BI users)

Key Sizing Assumptions:

- Databricks DBU rates in Central India (Pune) with applied regional multiplier.
- EXCLUDES Azure VM Infrastructure costs (VM instance pricing billed separately by Azure).
- ADLS Gen2 storage estimated at 7500 GB (₹12,750/mo).
- SQL Warehouse auto-stop configured at 10-15 minutes of inactivity.



Side-by-Side Architectural Matrix

Fabric Score: 96/100

Databricks Score: 87/100

ARCHITECTURAL DIMENSION	MICROSOFT FABRIC	AZURE DATABRICKS	ADVANTAGE
Cost & Licensing	Optimal ₹4,88,750 – ₹9,69,000/mo (Unified F-SKU)	Moderate ₹55,250 – ₹80,750/mo (DBUs + VM)	DATABRICKS
Scalability	Moderate Auto-smoothing CU capacity (F2 to F512)	Optimal Extreme horizontal cluster autoscaling & multi-cloud	DATABRICKS
Power BI Integration	Optimal Native Direct Lake zero-copy reporting	Complex SQL Warehouse connector (import/DirectQuery)	FABRIC
Data Engineering	Moderate Data Factory Pipelines & Synapse Notebooks	Optimal Delta Live Tables (DLT) & PySpark native	DATABRICKS
Machine Learning	Complex Synapse ML & AutoML (standard)	Optimal Native MLflow, Feature Store, GPU clusters	DATABRICKS
Streaming Processing	Moderate Eventstream & Real-Time Intelligence	Optimal Structured Streaming & Delta Engine	TIE
Data Governance	Optimal Microsoft Purview & Workspace roles	Optimal Unity Catalog (Cross-cloud fine-grained)	DATABRICKS
Operational Complexity	Optimal Low (Fully Managed SaaS)	Complex Medium-High (Cluster & Workspace Admin)	FABRIC
Team Learning Curve	Optimal Minimal (SQL & Power BI native)	Complex Moderate (PySpark & Delta Lake ramp-up)	FABRIC
Time to Production	Optimal 2–4 weeks (Turnkey SaaS workspace)	Moderate 4–8 weeks (Workspace & cluster setup)	FABRIC

↘ Cost Optimization & Best Practices

MICROSOFT FABRIC

Save up to ₹1,87,000/month (~38%)

Switch from Pay-As-You-Go to 1-Year Capacity Reservation

Commit to a 1-year Fabric Capacity reservation for baseline F-SKU workloads.

💡 Actionable Steps:

- ✓ Evaluate 30-day peak Capacity Unit (CU) utilization in Fabric Capacity Metrics app.
- ✓ Purchase 1-year reserved capacity for baseline F-SKU in Azure Portal.
- ✓ Keep pay-as-you-go enabled for seasonal burst capacity.

AZURE DATABRICKS

Save up to ₹21,250/month (~35%)

Configure Auto-Stop & Utilize Spot VM Instances for Batch Jobs

Set aggressive auto-stop timeouts on SQL Warehouses and configure Spot VMs for non-critical pipelines.

💡 Actionable Steps:

- ✓ Set SQL Warehouse auto-stop to 10 minutes of inactivity.
- ✓ Enable Spot VMs with fallback to On-Demand for daily ETL Jobs Clusters.
- ✓ Migrate ad-hoc analysis notebooks to Serverless Compute to pay strictly for query duration.

BOTH

Reduced memory overhead & zero data movement

Adopt Direct Lake & Delta Parquet Storage Standards

Store all silver/gold data in open Delta Lake format to eliminate ingestion hops and import mode refreshes.

💡 Actionable Steps:

- ✓ Standardize all data transformations on Delta Parquet format.
- ✓ Use OneLake shortcuts or Unity Catalog external locations for zero-copy access.
- ✓ Enable Direct Lake mode on Power BI semantic models for instant report performance.

📘 **Estimated using publicly available Microsoft and Databricks list pricing as of July 2025.** All cost figures are directional estimates for **Central India (Pune)**. Actual costs depend on region, workload characteristics, infrastructure choices, autoscaling, reserved pricing, enterprise agreements, and future pricing updates. **Verify estimates before making purchasing decisions.**

India Pricing Note: Fabric Central/South/West India rate confirmed at \$0.24/CU-hr vs \$0.18 in US East (~33% premium). Databricks DBU rates are region-stable, but underlying Azure Central India VMs run ~15-25% above US East.